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Using morphological and functional polycentricity analyses to study the Indonesian urban spatial structure: the case of Medan, Jakarta, and Denpasar

Erie Sadewo^{a,b}, Ibnu Syabri^a, Anzhelika Antipova^c, Pradono^a and Delik Hudalah^a

^aSchool of Architecture, Planning, and Policy Development, Institut Teknologi Bandung, Bandung, Indonesia;

^bBadan Pusat Statistik, Jakarta, Indonesia; ^cDepartment of Earth Science, The University of Memphis, Memphis, TN, USA

ABSTRACT

The existence of polycentric urban form has been widely discovered in worldwide urban development, yet the topic has received little attention in Southeast Asia. This study aims to explore urban spatial development in Indonesia, focusing on three historically globalized regions: the metropolitan areas of Medan, Jakarta, and Denpasar. It comprises subcenter identification and comparative morphological and functional polycentricity analyses using the latest economic census and commuter surveys data. The study confirms the polycentric structure of the Jakarta Metropolitan Area, while the other two metropolitan areas have a monocentric structure. The structural difference is due to the process of industrialization that shaped the development of metropolitan areas in Indonesia in various ways regardless of the spatial planning strategies.

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Introduction

A polycentric structure offers benefits for a more desirable form of urban development. Its proponents argue that polycentricity produces a better balance between the positive and negative aspects of agglomeration (Faludi 2004; Meijers 2007; Parr 2004). Polycentric metropolitan areas are relatively larger than monocentric ones and have a higher population density, greater per capita income, lower poverty rates (Arribas-Bel and Sanz-Gracia 2014), and higher labor productivity (Meijers and Burger 2010; Veneri and Burgalassi 2012). Moreover, the polycentric urban structure could help to increase individual travel pattern efficiency (Li et al. 2018; Lin, Allan, and Cui 2015). Much evidence has shown that polycentricity could reduce exposure to risk and enhance urban resilience (Sharifi 2019). Therefore, the study of polycentricity has received great attention through theoretical (see Wheaton 1979; Griffith 1981; Fujita and Ogawa 1982; Berry and Kim 1993; Lucas and Rossi-Hansberg 2002) and empirical explorations (i.e. McDonald 1987; Giuliano and Small 1991; McDonald and Prather 1994; McMillen and McDonald 1998; Antipova and Ozdenerol 2013; Arribas-Bel and Sanz-Gracia 2014; Antipova 2015; Huang, Liu, and Zhao 2015; Antipova, Skryzhevskaya, and Kung 2017).

Despite these positive features, other researchers have linked polycentricity to a “way of seeing” urban space and as a desired development strategy that promotes social, economic, and environmental goals to problems and limitation. Critics point out that higher GDP per capita in polycentric countries is more likely to be achieved on the national rather than regional level (Brezzi and Veneri 2015), which explains the increasing disparities between regions (Meijers and Sandberg 2008; Veneri and Burgalassi 2012). Polycentricity causes lower levels of urban productivity, although this impact appears to weaken as a city’s population increases (Li and Liu 2018). Meanwhile, the effect of the polycentric structure in reducing urban traffic congestion in China weakens as the metropolitan population size exceeds a certain threshold (Li, Xiong, and Wang 2019). The benefit of short commutes in polycentric cities could only be realized through a better balance between jobs and resident workers in each population center (Li et al. 2018). Thus, the effectiveness of the polycentric urban structure in reducing traffic and promoting sustainable development depends on many aspects.

Regardless of these mixed findings, the polycentricity discourse tends to shift from an analytical tool into a normative agenda in the form of a spatial policy (Davoudi 2003). Various places in Europe, the United States, and Asia have applied normative practices of polycentricity. In Europe, the concept has been adopted at the regional and national scale (Meijers 2008). Several locations in the United States have implemented the policy of polycentricity at the metropolitan scale, including Portland, Sacramento, Washington, DC, and Chicago. The adoption of this spatial structure is reflected in the long-term plans prescribing a polycentric hierarchy for these metropolitan areas (Knaap et al. 2016). Since the early 1980s, China made attempts to implement a “decentralized cluster style” (*fensan zutuan shi*) of development in Beijing through comprehensive city planning (Huang, Liu, and Zhao 2015, 11,633). However, the success of these efforts has been limited despite changes in Beijing’s spatial structure, including its residential suburbanization and development of several subcenters in the suburbs. Notably, the overall spatial distribution of employment still largely follows a monocentric form with jobs concentrated in the city center accounting for almost 60% of the city’s total employment.

Recent studies found evidence that the polycentric structure is an emerging urban development trend in East Asia (Phelps and Ohashi 2018). However, little is known about the dynamics of polycentricity in the context of Southeast Asia. While Southeast Asian cities have been recognized as being different from industrialized Western cities, they have a similar historical engagement with global capital flows. For example, the Netherlands East Indies was a destination for foreign investment since the end of the nineteenth century and produced a vast amount of wealth for the Dutch colonial government (Gordon 2014). Interaction between cities in the Southeast Asian region and the more advanced cities in the western countries, those so-called “metropolitan countries”, resulted in different phases of urban development with both convergence towards and divergence from the western model (Dick and Rimmer 1998, 2306). Since the 1970s, academic views of urban development in Southeast Asia shifted from the third world cities to world cities, to desakota, and lastly volatile globalization (Rimmer and Dick 2018, 11). Consequently, the current urban form is inseparable from its global context. However, contemporary studies have failed to address the causes and consequences of polycentric urban spatial development in Southeast Asia as their focus was on the transition of human settlements rather than to capital accumulation and the system of production.

This paper explores the current metropolitan spatial development in Indonesia. The Indonesian spatial pattern of urbanization reflects regional disparity and the country's economic development policy is biased towards urban areas (Firman 2004). The resulting combination of high population density and lagging investments in infrastructure has led to congestion, pollution, and an increased risk of disasters in many urban areas (World Bank 2016). A better understanding of urban spatial organization might contribute to appropriate land planning strategies to address these issues (Antipova and Ozdenerol 2013). In this respect, several attempts have been made to identify the metropolitan spatial structure of the Jakarta Metropolitan Area (JMA), resulting in different conclusions. While Hakim and Parolin (2009) found that the JMA's spatial structure matches neither the monocentric nor the polycentric model, Hudalah et al. (2013) concluded that it is shifting from a scattered spatial arrangement towards a polycentric region. Sadewo, Syabri, and Pradono (2018) supported the latter argument, emphasizing that the JMA's spatial form has transformed from a multifunctional to a mono-functional polycentric structure. The focus of these studies was limited to the JMA, while the contrasting conclusions primarily resulted from their different ways of measuring polycentricity as none of those studies used a combination of morphological and functional aspects in analyzing the urban spatial form.

For Indraprahasta and Derudder (2019), a long-term evolutionary process with different phases has shaped the JMA's urban character and function in the global economy. Nevertheless, only little is known about the relationship between the economic function and the urban form of the JMA and other metropolitan areas in Indonesia. In polycentric urban regions, key facets and dimensions such as internal dynamics, territorial context and path-dependencies should be considered as the source of development variations (Schmitt et al. 2015). By comparing the strikingly distinct Medan Metropolitan Area (MMA) and Denpasar Metropolitan Area (DMA) to JMA, this study argues that contemporary urban form is also path-dependent to the contextual development base. Thus, this study uses Robinson's (2011) *individualizing strategies* rather than applying a universalizing comparison approach to compare different metropolises. It discusses the historical aspects of each metropolitan area to explain the differentiated process leading towards contemporary urban form.

This objective is achieved in three ways, i.e. identify the subcenters for each metropolitan area, compare the level of polycentricity in those areas, and discuss how different growth trajectories within a similar planning system could shape urban spatial development in unique ways. The paper contributes to the body of literature by providing rare empirical evidence of urban polycentric development in the Southeast Asian context.

The paper comprises seven parts. The introduction provides the research background and the second section gives a brief overview of the study areas. The third section reviews previous studies on urban subcenter identification and polycentric measurement, while the fourth section discusses data and methodology. The fifth section presents the main findings, the sixth section discusses the historical and contextual development of the case studies, while the last section contains the conclusions.

An overview of the case studies areas

The Indonesian government has designated 14 metropolitan development areas in its 2015–2019 National Mid-Term Development Plan (RPJMN). In 2017, these areas were

home to almost 34% of the national population, with an annual growth average of 1.68% between 2010 and 2018. These metropolises are unevenly distributed: three located in Sumatra, five in Java, one in Kalimantan, two in Sulawesi, and two in Bali and Nusa Tenggara. The examination of location quotient (LQ) and export value of 14 metropolitan areas in Indonesia has revealed that the MMA has a plantation economic base and is the largest metropolitan area in terms of agricultural export value. The JMA's economy is strongly supported by commercial and business activities, although their largest export sector is manufacturing (Table 1). Meanwhile, the DMA has developed global-scale tourism-related service activities. Thus, we selected the MMA, the JMA, and the DMA as the case studies because these metropolitan areas represented the strongest linkage to the global network of capital accumulation in the sectors of agriculture, manufacturing, and service, respectively.

The MMA is located on the northeastern coast of Sumatra, with easy access to the busy Malacca Strait shipping lanes and closeness to several neighboring countries (Figure 1). Owing to this strategic geographic location, it plays an important role in the Indonesia-Malaysia-Thailand Growth Triangle. The MMA consists of 52 sub-districts covering an area of 3016.97 km². It is also known as Mebidangro, abbreviated from the names of the places that compose the area: Medan City as its metropolitan core is surrounded by Binjai City, Deli Serdang Regency, and parts of Karo Regency. Medan is the capital of North Sumatra Province, the fourth most populous city in Indonesia, and the largest city outside Java (Tarigan et al. 2017). Karo Regency was excluded from the study as it is located farthest from the metropolitan core and only parts of it are included in the MMA. Therefore, in this paper, the term MMA refers to Mebidang rather than to the entire Mebidangro region.

Located on the northwestern coast of Java, the JMA comprises 183 sub-districts across the three provinces of West Java, Banten and the special province of DKI Jakarta. DKI Jakarta is the metropolitan core and is surrounded by the peripheral municipalities of Bogor, Depok, Bekasi, and Tangerang. The JMA is also known as Jabodetabek, a combination of the initial letters of each city it is composed of. Serving as the national capital, the JMA plays a prominent role in the development of Indonesia in many aspects, i.e. social, economic, and political (Rustiadi et al. 2015). As the main gateway for passengers and goods into Indonesia, it has experienced rapid urbanization and evolved into an increasingly multicultural and cosmopolitan area. With an area of nearly 6392 km² and a population of 30.97 million in 2015, the JMA is the largest metropolitan area in Indonesia, and the second largest in the world (Sadewo, Syabri, and Pradono 2018). The region contributes 25% of the national GDP, mainly from manufacturing and high-order services activities.

The third study area represents an example of small island development. The DMA is composed of 27 sub-districts and four regency-level administrative areas, covering 1753.63 km². It comprises the core city of Denpasar and its surrounding Badung, Gianyar and Tabanan Regencies, and is abbreviated as Sarbagita. Strategically located in the southern part of Bali Island, the metropolitan area is renowned for its strong cultural identity and a beautiful landscape that provides the basis for tourism. It has a population of 2.5 million and is ranked third for annual population growth between 2010 and 2018 after the JMA and the Banjarmasin Metropolitan Areas.

Table 1. Selected economic indicators of 14 metropolitan area in Indonesia, 2016.

Region	Metropolitan Area	GDP (Bio Rp)	Highest LQ Value	Basis Sector with Highest LQ Value	Biggest Regional Export Commodity	Regional Export Value (Mio US\$)	Contribution to Total Regional Export Value (%)
North Sumatra	Medan MA	297,007	1.7	Agriculture	Palm oil	4346.9	56.0
West Sumatra	Padang MA	70,822	5.1	Mining & Quarrying	Palm oil	1536.0	89.9
South Sumatra	Palembang MA	175,548	3.4	Agriculture	Rubber	1412.7	71.4
Jakarta, West Java, & Banten	Jakarta MA	3,091,373	1.6	Establishment & Business activities	Electrical/ machinery equipment	10,177.8	21.9
West Java	Bandung MA	373,038	1.5	Transportation & storage	Electrical/ machinery equipment	6569.6	25.5
Central Java	Semarang MA	273,358	2.3	Agriculture, forestry, & fisheries	Yarn, textile product	3292.5	61.1
	Surakarta MA	206,646	3.5				
Special Region of Yogyakarta	Yogyakarta MA	86,765	2.2	Accommodation & Restaurant	Non-knitted apparel & clothing accessories	166.0	49.7
East Java	Surabaya MA	841,608	2.0	Mining & Quarrying	Jewelry & other precious metal	4407.7	23.3
Bali	Denpasar MA	130,750	5.1	Accommodation & food service	Fish & shrimp	188.5	37.3
West Nusa Tenggara	Mataram MA	60,573	5.3	Mining & Quarrying	Ores, slag & ash	1581.7	99.7
South Kalimantan	Banjarmasin MA	64,051	10.8	Mining & Quarrying	Coal & lignite	5359.3	94.0
South Sulawesi	Makassar MA	168,795	2.0	Water supply, sewage, waste management & remediation activities	Nickel	988.9	70.2
North Sulawesi	Manado MA	51,420	3.0	Mining & Quarrying	Copra	714.5	84.2

Source: author elaboration from 2016s regional GDP and Export data (BPS).

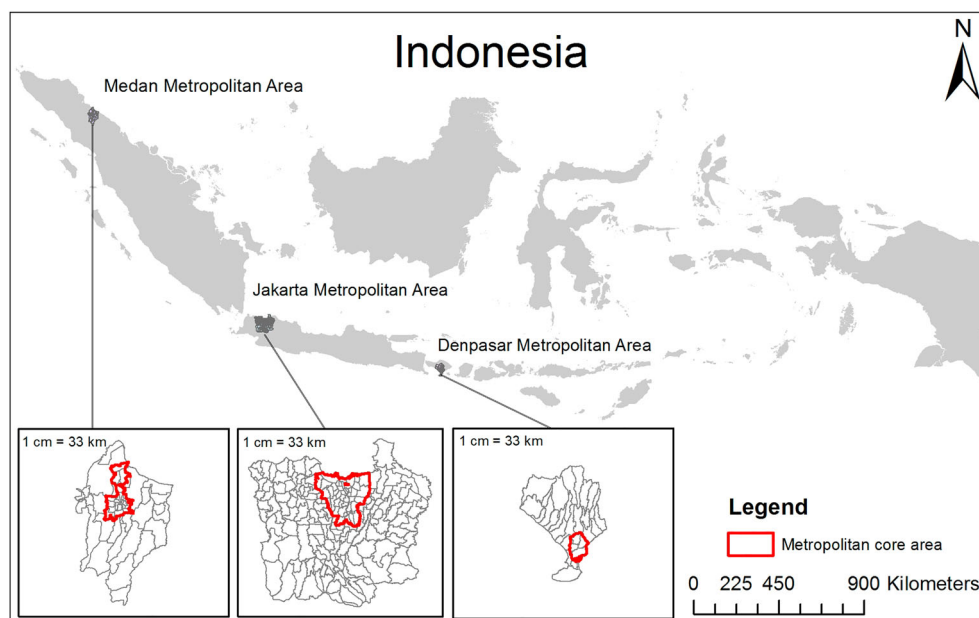


Figure 1. The Location of the study areas.

Identification of subcenters and measuring polycentricity

Studies on polycentricity have used multiple and complex perspectives. A polycentric urban structure is formed when one or more employment subcenters develop beyond the Central Business District (CBD) (McMillen 2001, 448). An example of a polycentric urban form is a metropolis consisting of several smaller, specific, and relatively adjacent activity centers, as a natural form generated by the evolutionary process of a city (Hall and Pain 2006). It is driven by the decentralization of economic activities, increased mobility, complex cross-commuting, and fragmented spatial distribution of activities (Davoudi 2003). Focused on intra-metropolitan areas, Hall (1999, 178) classified the polycentric urban structure into six main activity centers: a traditional business core; secondary business core; tertiary business core or inner edge city; outer edge city; outermost city; and scattered specialized concentration of activities. Yet, the concept can also be applied at larger different geographic scales such as the inter-urban and inter-regional scales (Davoudi 2003).

Polycentric models of urban development have been developed to address the deficit of the monocentric model in adequately explaining the expanding urban land use, which encourages the formation of multiple smaller centers within urban commuting areas. Built upon empirical findings, the mainstream research on polycentricity regards these subcenters as major employment locations (Alegria 2016). Cladera, Marmolejo Duarte, and Moix (2009) classified the employment density-based subcenter identification methodology into five main groups: identification of local peaks in employment density; use of thresholds reference or cut-off values in the densities and absolute employment values; parametric methods that analyze residuals in a regression equation of employment density; nonparametric methods commonly using locally weighted regression models to

identify local employment density peaks; and some other variants such as a cubic spline density function. However, several other methods can also identify subcenters, such as proportional approaches and spatial statistics (Alidadi and Dadashpoor 2017).

The employment density peaks (EDP) approach identifies subcenters as job locations of a certain density (McDonald and McMillen 1990). This method is criticized for its lack of theoretical foundation and objectivity. Meanwhile, the proportional approach compares the absolute number of workers in a location to the entire metropolitan area (Alegría 2016; Shearmur and Coffey 2002). It uses the employment-to-residential workers ratio (E/R ratio) to estimate the net in-commuting and employment attraction. Both methods use arbitrary measures but the latter is considered more accurate. The EDP approach is also considered to be static and incapable of showing economic agglomeration because it excludes the distance between firms. The threshold reference approach, which is based on the number of jobs per acre and the absolute number of employment (Giuliano and Small 1991), is a combination of the EDP and proportional methods. Coffey and Shearmur (2001) have shown that the use of a combination of approaches produces more efficient measurements.

The identification of subcenters in parametric and nonparametric methods is based on the residual analysis of the density function. Both methods compare the residuals between real and predicted values and consider observations with the residuals that are statistically different from zero as candidate subcenters. The parametric method is simpler, yet it provides rigorous results. However, the parametric method only portrays the metropolitan area one-dimensionally based on the assessed location distance to the CBD, while disregarding the complex three-dimensional geographical nature of urban systems, which can be better understood by analyzing historical development (Cladera, Marmolejo Duarte, and Moix 2009, 2843). A nonparametric approach improves the parametric model's efficiency in identifying local maximums of employment density as a result of historical urban development processes. Several nonparametric models have been proposed, such as a two-stage procedure involving a smoothed employment density function in the first stage and a semiparametric estimation in the second stage (McMillen 2001), and quantile smoothing splines (Cladera, Marmolejo Duarte, and Moix 2009). Meanwhile, the Exploratory Spatial Data Analysis (ESDA) approach identifies subcenters by considering spatial autocorrelation and spatial heterogeneity (Baumont, Ertur, and Gallo 2004; Hakim and Parolin 2009). The advantages of the method include flexibility in scaling, avoidance of arbitrary cut-off values, and ease of operation, making it preferable when studying a large number of areas (Arribas-Bel and Sanz-Gracia 2014; Li and Liu 2018; Li, Xiong, and Wang 2019). Nevertheless, the use of this method requires complete supporting data such as maps and aggregate location data. Another disadvantage of the nonparametric method is the difficulty to determine whether the identified "peaks" are true subcenters that correspond to the spatial reality or random peaks in the metropolitan employment pattern (Cladera, Marmolejo Duarte, and Moix 2009, 2844).

While many statistical approaches are more appropriate for the analysis of individual urban areas, most comparative urban studies use index values. Comparative polycentricity considers the morphological aspect of the urban system and the interrelated relationships between urban centers (Green 2007; Meijers 2008). The polycentricity does not only refer to urban center plurality of a given delineation of the physical dimensions of the urban system, but is also closely associated with the balance of urban center distribution

(Burger and Meijers 2012). Nevertheless, the polycentric urban region should fulfill not only the condition of separation between centers, but also the distribution, specialization, and interaction between them (Parr 2004). Thus, the subcenters in a balanced multi-urban center system should be functionally inter-connected (Green 2007). Empirical evidence has shown that the differences between morphological and functional polycentricity could occur to some degree (Burger and Meijers 2012). Hence, it is important to consider both aspects in spatial structure measurements of different regions for comparison.

Meijers and Sandberg (2008) suggested that an area should not be seen as monocentric or polycentric per se. For comparative purposes, polycentricity should be measured by some value ranging from fully monocentric to fully polycentric. Several frameworks have been developed to measure morphological polycentricity, whereas obtaining the value of the rank-size distribution from a regression slope is the most popular (Burgalassi 2010; Burger and Meijers 2012; Veneri and Burgalassi 2012; Li and Phelps 2018). This method is practical and straightforward (Meijers 2008), and provides accurate information about the hierarchy of a center as a proxy of each city's importance (Parr 2004).

From the functional perspective, a polycentric structure requires a certain density and a balance of interaction flows between its nodes of activity. Theoretical discourses suggest that a close relationship between commuting patterns and urban structures leads to polycentricity (Gordon, Kumar, and Richardson 1989; Helminen et al. 2012; Lin, Allan, and Cui 2015). Therefore, studies widely use commuting patterns to assess the functional aspect of polycentrism, although the functional relation could appear in various forms, from physical flows of goods to informational exchange. Green (2007) proposed social network analysis (SNA) to identify the two requirements of functional polycentricity. This approach is versatile and scalable, and applicable from individual to regional to national levels. It allows to compare polycentricity functionally across scales and among different functions. The method has been used not only for identification of functional polycentricity (Liu, Derudder, and Wu 2016; Mu and Yeh 2016), but also for comparison of morphological polycentricity (Hall and Pain 2006; Vasanen 2013; Liu et al. 2019; Yue et al. 2019). However, it still has drawbacks of determining the maximum value limits, which has an element of subjectivity.

Methods and data

This study adopts the identification of subcenters approach following the classical monocentric standard density function hypothesis, which mainly comes in negative exponential forms, to compare the polycentricity of the MMA, the JMA and the DMA. In this regard, this study applies a conventional statistical approach using parametric modeling. This involves estimating parameters generated by the models of each sub-district's employment density, as a function of accessibility to the CBD of each metropolitan area. However, since the data of the actual distance between sub-districts were not available, the Euclidean distance between each sub-district centroid is used as a proxy. We obtained the distances to the CBD in two stages. First, we arbitrarily selected CBD points within the traditional city center known as "*alun-alun*" in Indonesia. An *alun-alun* in modern-day Indonesian cities represents a large, central square used for recreation. It is typically surrounded by important buildings such as mayor's houses, offices, and a central mosque. Specifically, we used Merdeka Square in West Medan sub-district in the MMA, Monas Square in Gambir in the

JMA, and Puputan-Badung Square in West Denpasar in the DMA to identify the CBD location. Second, we determined a centroid point for each sub-district. Then, we calculated the distance between each centroid to the traditional CBD as a straight-line distance.

Using the parametric model also allows for a comparison of the numbers and location of subcenters obtained by previous studies. To relax a normality assumption and obtain the best model, we tested five parametric models using different functions of employment density: a linear model (LIN), a negative exponential model (EXP), a logarithmic version of the negative exponential model (also called a standard model, STD), an inverse potential or a gravitational model (GRAV), and a logarithmic version of the gravitational model (Log-Log). We selected the best model by evaluating its coefficient of determination (R^2) as a relative measure of fit. The spatial symmetry breaking by creating disaggregated parametric models following the major transportation axes was also considered if the initial result is dissatisfactory. For a more detailed discussion of the parametric methods used for subcenter identification, the reader may refer to Cladera, Marmolejo Duarte, and Moix (2009). Since some models could produce similar R^2 values, the Root Mean Square Error (RMSE) as an absolute measure of fit was also employed for best model selection. We obtained the subcenter based on the value of the positive residuals from the best model with the best goodness-of-fit that exceed a threshold of employment density. Giuliano and Small (1991) proposed a threshold of 1000 jobs or a minimum density of 5,000 jobs per square mile for this threshold. However, we relaxed the threshold in this study to 1000 employment/km² because we limit our analysis in this study to Medium and Large Establishment (MLE) data for sub-districts.

The decision to limit our analysis to MLEs was to compare our findings with the earlier research (see Hudalah et al. 2013 and Sadewo, Syabri, and Pradono 2018). This study uses similar aggregate data at the sub-district level as Hudalah et al. (2013). Nevertheless, this study uses complete non-agricultural MLE activity data from the 2016 Economic Census, as in Sadewo, Syabri, and Pradono (2018), to avoid reaching a partial conclusion which may result from the assessment of the spatial distribution of a single type of economic activities such as manufacturing (Hudalah et al. 2013). The use of MLE data is statistically representative because, despite being small, non-agricultural MLEs contribute significantly to employment. To illustrate, in 2016, 2.61% of MLEs accounted for 30.92% of the total employment in the MMA, while 4.61% of MLEs accounted for 47.64% of the employment in the JMA, and 4.09% of MLEs represented 34.94% of the employment in the DMA (see Table 2).

There is a large body of previous research comparing polycentricity between regions, but only a few studies use a combination of morphological and functional approaches (Burgalassi 2010; Cladera, Marmolejo Duarte, and Moix 2009; Hall and Pain 2006; Vasanen 2013; Liu et al. 2019; Yue et al. 2019). This study uses both approaches to obtain a more comprehensive conclusion. For morphological identification, this study follows Burger and Meijers (2012) to estimate the regression slope of rank-size distribution. Here, the size refers to the number of employment in the delineated municipality/regency and the nodes refer to the center of each municipality or regency within the metropolitan area. Consequently, there would be three urban regions with contrasting node numbers. Since including a greater number of nodes tends to produce a greater polycentric value, we chose a sample of a fixed number of nodes to estimate the slope (Meijers and Burger 2010). We created a regression model where we used a different number (2, 3,

Table 2. Some characteristics comparison between MMA, JMA, and DMA.

Characteristics	MMA	JMA	DMA
Number of administrative areas	3	9	4
Area (km ²)	2853.15	6935.19	1753.63
Population (000)*	4239	30,970	2428
Average density / Km ²	1486.05	4465.71	1384.59
Working population (%)*	39.77	45.09	55.82
Number of non-agricultural economic activities (000)	397	2.801	289
Share of MLE unit (%)**	2.61	4.61	4.09
share of MLE employment (%)**	30.92	47.64	34.94

Source: Badan Pusat Statistik.

*Data of 2014 for JMA and 2015 for MMA and DMA.

**Data of 2016.

and 4) of places in each region with the biggest employment size and regressed these places on their rank. Then we used the average value of each slope as the final morphological polycentric value. A flatter slope, or a value closer to 0 indicates a more polycentric metropolitan area. We adjusted the number of places in the MMA by splitting the Deli Serdang Regency into two sections because MMA had the smallest number of nodes. We set the sub-districts of areas mostly located east of 98,65° longitude as Eastern (E) Deli Serdang and their counterparts as Western (W) Deli Serdang. This division creates 12 equal sub-districts in both areas.

To identify the functional aspect of polycentricity, we used Green's (2007) SNA approach by comparing the General Functional Polycentricity (Pgf) index of the study areas with other Pgf index benchmarks. While the index value is theoretically between 0 and 1, empirical evidence from Europe shows that a functionally notable polycentric area has a maximum index value of 0.19 (Hall and Pain 2006, 52). We derived the data used in this method from the result of commuter surveys, published by Indonesia's Central Statistics Agency (BPS). The surveys were conducted in two phases: in the JMA in 2014, and in the MMA and DMA in 2015. The commuter survey in the JMA covers 13,120 household samples, while surveys conducted in the MMA and DMA comprise 2,400 household samples each. This survey uses a slightly different definition of commuting. The United States and several continental European countries define commuting as a recurring regular travel activity for the purposes of work or school, regardless of its destination. Meanwhile, commuting in Indonesia and several Scandinavian countries is defined as a daily travel activity that passes the municipal administrative boundary of their residential area. Thus, the location of employment which we used to compose the origin-destination (OD) matrix is based on the regency-level boundaries.

Findings

Subcenters in urban development

To identify the subcenters in each metropolitan area, we tested different parametric models of employment density as a function of accessibility to the CBD (Table 3). The models reveal a significant relationship between employment density in each sub-district and the distance to the CBD. We found that the exponential and standard models outperform others. In the JMA, the exponential model has a slightly higher coefficient of determination value ($R^2 = 55.39$) compared to the standard model ($R^2 = 54.55$). Nevertheless,

Table 3. Model of employment density as a function of the accessibility of the CBD.

Location	Statistics	LIN	EXP	STD	GRAV	Log-log
MMA	Constant	1573	–	8.133	–	6.743
	<i>a</i>	–	3.79E + 03	–	1084	–
	<i>b</i>	–54.68	–0.1797	0.2462	–0.1921	–0.9768
	<i>R</i> ²	30.47	56.32	59.26	23.46	24.99
	SSE	4.61E + 07	2.89E + 07	1.76E + 02	5.07E + 07	3.23E + 02
	RMSE	979.6	776.5	1.95	1028	2.65
	Sub-center	8	7	10	9	9
JMA	Constant	4.5E + 05	–	902.93	–	1323
	<i>a</i>	–	1.1E + 06	–	3.4E + 05	–
	<i>b</i>	–10,030	–11.19	12.02	–22.85	–242.3
	<i>R</i> ²	32.68	55.39	54.55	23.1	47.54
	SSE	9.9E + 10	6.5E + 10	5.7E + 04	1.1E + 11	6.6E + 04
	RMSE	233,600	190,200	178.7	249,700	191.3
	Sub-center	32	42	52	47	46
DMA	Constant	1242	–	7.77	–	5.60
	<i>a</i>	–	1933	–	578	–
	<i>b</i>	–42.72	–0.10	0.19	0	–0.63
	<i>R</i> ²	28.68	33.84	84.26	11.34	29.95
	SSE	1.45E + 07	1.35E + 07	21.57	1.80E + 07	95.98
	RMSE	761.40	733.40	0.93	849	1.96
	Sub-center	2	1	1	1	1

the Root Mean Square Error of the standard model is much better than that of the exponential model (178.8 compared to 190,200). Therefore, we selected the standard model as the best for the JMA. In the MMA and the DMA, the standard model resulted in better values for both the coefficient of determination and Root Mean Square Error. Furthermore, the standard model produced a superior coefficient of determination value ($R^2 = 84.26$) compared to other models in the DMA. This finding contrasts with Cladera, Marmolejo Duarte, and Moix (2009) who highlighted the superiority of the exponential and gravitational models. A high determinant coefficient implies that there is no need to break the symmetry.

Overall, the standard model identifies more subcenters compared to the exponential model and has a steeper curve slope, which indicates faster density decay at a relatively closer distance than the exponential model. Figure 2(a,b) shows that the MMA has 10 sub-districts as economic centers, all located within 10 km from the CBD and inside the Medan City core area. Thus, the extended CBD area in the MMA is likely to be a spillover rather than an independently developed subcenter. Yet, the DMA shows a different situation as the standard model could only identify Kuta sub-district as a primary employment center, contiguously neighboring the traditional CBD in West Denpasar (Figure 2(f)). Clearly, the CBDs still dominate the economies of the MMA and DMA. Thus, parametric method with standard model could capture the centrality tendency better.

In contrast, the standard model identified 52 sub-districts as economic centers in the JMA. The metropolitan area has at least four subcenter clusters, i.e. one inside the core, and the rest located 20–60 km from the CBD. The subcenters have developed near the metropolitan edge, most notably the Cikarang corridor in the east, Bogor City in the south, and the Tangerang-Cikupa-Balaraja corridor in the west (Figure 2(d)). The models clearly revealed the multi-subcenter nature of the JMA. Although this finding confirms Hudalah et al. (2013) and Sadewo, Syabri, and Pradono (2018), this study offers an important improvement, i.e. the parametric method in this study identified

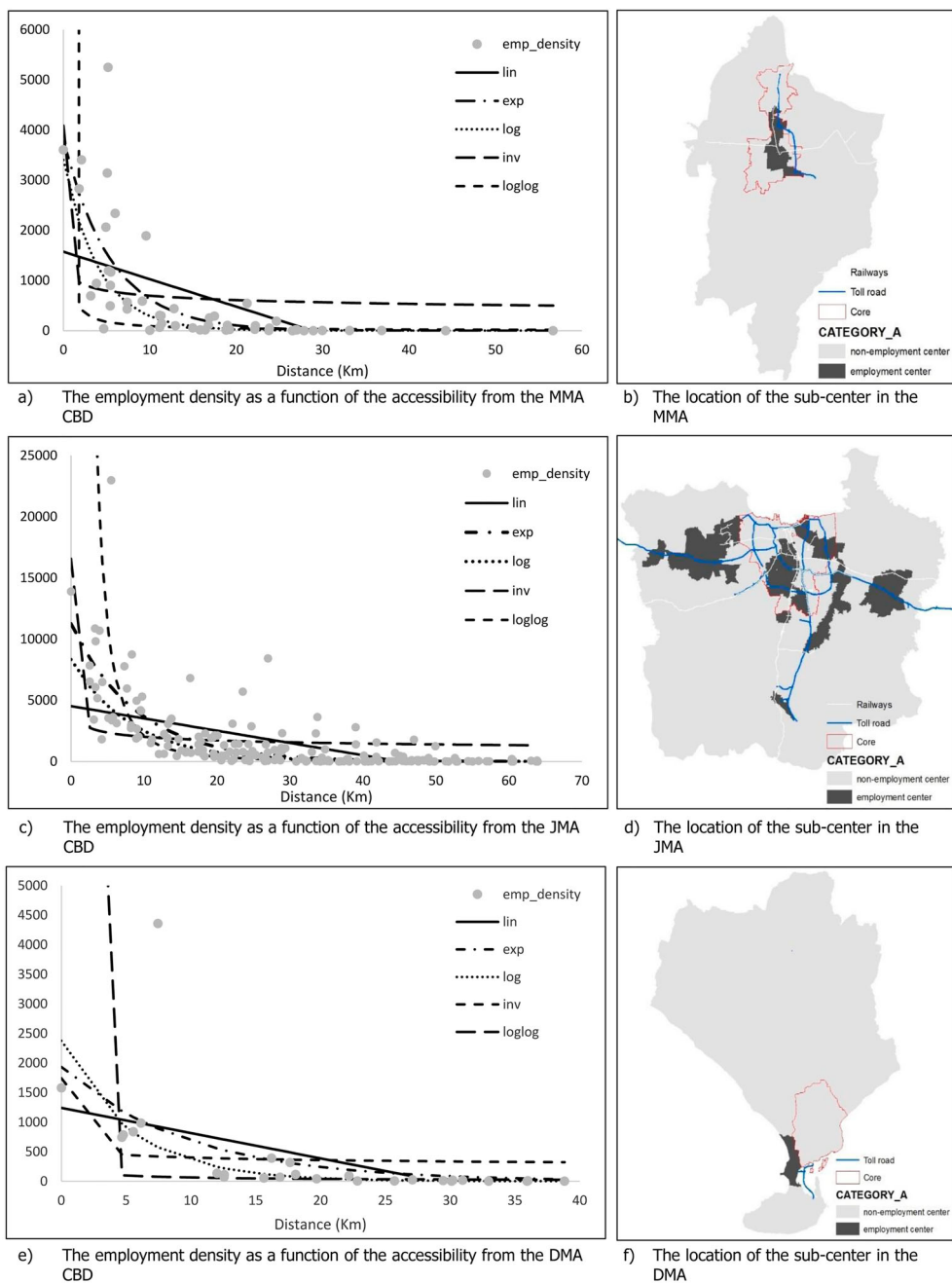


Figure 2. Spatial structure comparison between MMA, JMA, and DMA.

Bogor City in the south of the JMA as a subcenter while the above-mentioned studies failed to do so.

The use of a parametric method is more appropriate compared to the earlier studies of polycentricity in the JMA by Hudalah et al. (2013) and Sadebo, Syabri, and Pradono (2018) for two reasons. First, this study uses all non-agricultural MLEs, while Hudalah

et al. (2013) used manufacturing industries MLEs only. Economic Census data of 2016 revealed that manufacturing industries represent up to 70% of employment in each of the identified subcenters (the regencies of Bogor, Bekasi and Tangerang, and Tangerang City). However, Bogor City is different because the same data shows that the trade sector dominates its economy. Employment in the finance and insurance sectors ranks third in the economic structure of this historically established urbanized area. Such characteristics cannot be identified using only manufacturing-based MLE data.

Second, Sadewo, Syabri, and Pradono (2018) indicated that the spatial pattern of business establishments in the JMA developed along the toll road corridor connecting Jakarta (the CBD) with the metropolitan subcenters. The employment distribution decreases further south along the toll road towards Bogor City, implying a greater importance of distance from the CBD in the JMA's southern suburban development. However, the ESDA method used for subcenter identification makes no assumption of the influence of distance decay on employment density. While the employment density in the southern region of the JMA is lower, the distribution of Bogor City's employment density is only deviated slightly from the mean. This factor prevents the ESDA method from identifying Bogor City as a potential subcenter. This study, in contrast, uses the distance decay as a weight of the parametric method and can better capture the actual economic development patterns.

The comparative polycentricity

The location of employment centers within the three metropolises reflects their contemporary urban structure. With respect to the average rank-size regression slope of different numbers of incorporated places (2, 3, and 4), the three metropolises exhibit different spectrums of polycentricity. Based on the measurement used by Meijers and Sandberg (2008), a morphological value of -0.44 for the JMA suggests that this metropolis is very polycentric, as Tangerang City and Tangerang Regency, the two locations that ranked second and third in employment, are located outside the JMA's urban core. Meanwhile, the morphological values of -1.12 for the DMA and -2.21 for the MMA indicate average polycentric and very monocentric structures, respectively. The shift of the employment center to Badung Regency, just outside the traditional urban core, creates a more balanced morphology in the DMA. As for the strong monocentricity of the MMA, it can be mainly explained by the greater employment discrepancy between the metropolis's urban core and peripheral area. Thus, the differences in employment center location between the MMA and DMA have influenced its morphological polycentricity levels, regardless of their similar tendency of centrality.

We also identified polycentricity based on the functional connection between settlements from the intra-metropolitan commuting patterns. Having specified an administrative municipal boundary to limit settlement locations, we found that commuting activity makes up only a small part of daily commutes. Table 3 shows that the share of commuters in the three metropolitan areas only accounts for less than 20% of the total working population. Nevertheless, commuter flows can be quite significant if we would consider the absolute number of the labor force. In the JMA, for instance, the number of daily commuters reaches 3.5 million people. Compared to the MMA or the DMA, which have only 315 and 187 thousand commuters, respectively, the JMA has a more complex commuting

pattern. Although the traditional commuters from the suburbs to JMA's core still dominate the travel patterns, the flow to the suburban areas is sizeable. There were 10.2% reverse commuters and 33.39% cross commuting activities, which indicate the importance of suburban economic activities. Such complexities do not occur in the MMA, while in Denpasar, the reverse and cross commuting flows together make up a greater value of 59.89%. Nevertheless, this situation cannot be interpreted as an increase in complexity since the CBD is located in the suburban area contiguous to the DMA core (Figure 2(f)).

We verified the importance of suburban economic activity using the general functional polycentricity index (Pg_f) proposed by Green (2007). Although the special functional polycentricity index (Ps_f) of the MMA is quite balanced between in- and out-commuting, the overall Pg_f value was low compared to Hall and Pain's (2006, 52) benchmark. Table 4 shows an unbalanced Ps_f for the DMA, whereas its out-commuting flow is more polycentric than the in-commuting flow. Nevertheless, its Pg_f value is equal to that of the MMA. Thus, the MMA and DMA are not polycentric in terms of a functional network between settlements. The JMA, on the other hand, has a larger out-commuting Ps_f value but the daily travel pattern is spread more evenly between traditional and cross commuting, the resulting Pg_f value of 0.11 imply that the JMA currently has a medium level of functional polycentricity.

The contextual perspective

At some point, the contrast between the three metropolises urban forms may be largely determined by the striking difference in their population size. A review of the subcenter formation theory by McMillen and Smith (2003) emphasized the importance of population numbers and commuting costs in determining the urban spatial equilibrium. Nevertheless, previous studies have also shown that neither population nor urban size could guarantee polycentricity. To illustrate, Hong Kong was monocentric in terms of population concentration but polycentric when it comes to employment locations (Walls and Wang 1997). Griffith and Wong (2007) identified the New York Metropolitan Area, ranked first in population size in the United States, as monocentric. Meanwhile, they identified other metropolises such as Atlanta or Cleveland, which ranked 11th and 16th, respectively, as polycentric. Polycentric structures could also be found in almost all European urban regions, regardless of their size (Meijers, Hoogerbrugge, and Cardoso 2018). In China, most mega-city regions are surprisingly lacking in polycentricity (Mu and Yeh 2016). This suggests that in several cases, population size is not the primary determinant. Thus, we might need to consider other factors, since it becomes increasingly difficult to separate history and geography, time and space (Soja and Hadjimichalis 1979).

Table 4. Metropolitan functional network of MMA, JMA, and DMA.

Characteristics	MMA	JMA	DMA
Ratio of commuter to labor force numbers	0.18	0.19	0.14
Traditional commuting (%)	80.63	56.41	40.11
Reverse commuting (%)	16.83	10.20	41.71
Cross commuting (%)	2.54	33.39	18.18
Ps _f (In-commuting)	0.03	0.10	0.01
Ps _f (Out-commuting)	0.04	0.12	0.07
Pg _f	0.03	0.11	0.03

The MMA has grown with the development of Medan City, as a trading center for various plantation commodities produced in the surrounding areas such as rubber, tobacco, palm oil, tea, and coffee. The development of Medan begun in the nineteenth century, when the Sultan of Deli invited Dutch businessmen to open tobacco plantations and granted them concession rights (Elida 2018). The success of the tobacco plantation encouraged many entrepreneurs to expand the plantation business in the region to rubber and palm oil. Following the massive capital inflows, the Dutch moved their government center from Bengkalis to Medan (Nurhabsyah 2019). Supported by the capital and management system from the Netherlands and various European companies, a 200-kilometer rubber belt emerged during the first four decades of the 1900s, equipped by the Deli railway system (Kaur 2004, 86–87). The rapid growth of plantations by the early of the twentieth century increased the demand for labor. To fulfill this demand, the *Coolie Ordinance* policy by the Deli Planters Association (DPU) facilitated the migration of workers from Java, India and China, creating a multicultural society. However, many elements of the migrant worker's native cultures have not been practiced again today, so that outsiders tend to see the present population as a homogeneous identity (Nurhabsyah 2019).

The shift to rubber plantation of today North Sumatra Province began in the middle of the twentieth century in anticipation of the needs of the automotive industry in the western countries. To date, rubber product-related industries generate the largest employment and second-largest output value in the Province. Meanwhile, palm oil-related industries rank second in employment and first in total output share. Within the last two decades, there has been an increasing role of both commodities in global trade. Data from Indonesia's Central Statistics Agency (BPS) shows that in 1997, palm oil and rubber products from North Sumatra made up 45.7% of the region's total export. In 2017, the share of these commodities in the total export of North Sumatra had increased to 62.4% and one-fifth of the total national palm oil and rubber product exports. One of the biggest palm tree plantation concessions managed by the 2nd State-Owned Plantation Holding (PTPN 2), and located in the Lubukpakam Sub-district of Deli Serdang, has operated from the early 1970s. In 2016, palm oil products and its derivatives contributed nearly 73% of Deli Serdang's capital turnover.

In the post-independence era, the growth of Medan City led to changes in its spatial pattern (Tarigan et al. 2017). Subsequently, the city's economic focus shifted from the plantation-based primary economic activities to industry and services, reflected by the establishment of the first industrial estate with an area of 514 hectares in the southern part of Medan in 1988. To accommodate the changing nature of the economy, the municipal government has implemented multicentered spatial planning policies since 2011. However, these policies are still limited to Medan City. Under the spatial masterplan developed by the government of Medan City, the city is expected to have two services-based centers and eight subcenters with each having specific functions (*Medan City Spatial Planning 2011–2031*, 2011).

The JMA has been managed and contested by various forces over time until it came under the control of the Dutch East India Company (VOC) in 1619. The transition between ruling power from the VOC to the Dutch colonial government transformed the area from the company headquarters to a capital city. As such, Jakarta became the center of the Dutch colonial government in Indonesia, while its future expanding metropolitan area (the JMA) developed as a base for securing and expanding their commercial activities in Asia (Indraprahasta and Derudder 2019). The Indonesian government

repeated this strategy after the country's independence. Since the implementation of liberalization policies in the mid-1980s, the region has seen an increase in capital inflow, especially in the manufacturing sectors.

Industrialization is the key factor for the rapid urbanization in the region (Hudalah et al. 2013). Notably, the urban transformation of the JMA is driven mainly by economic expansion manifested in the rapid construction of industrial complexes and new satellite cities. This resulted in a spatial expansion with mixed land use on the outskirts of Jakarta (Rustiadi and Panuju 2002). To avoid social discomfort, gated communities developed rapidly along with various amenities, reflecting American-styled metropolitan expansion (Dick and Rimmer 1998). During the "New Order" era of 1966–1997 (named so to distinguish it from the economic collapse and political instability of the preceding "old order" period), Indonesia's development process was marked by the absence of planning and regulations (Rukmana 2015). Meanwhile, liberal economic policies resulted in planned as well as unplanned urban forms (Winarso 2011). This so-called urban duality was a consequence of the increased demand for land, which encouraged massive land banking practice by the private sector, causing largely uncontrolled development.

The DMA has maintained its status as the center of Bali Island's economic activities, the largest tourist destination in Indonesia and one of the most popular holiday locations in the world for many years. Over the last two decades, Bali has accounted for 32% of Indonesia's total foreign tourist visits. The tourism-related economy such as accommodation and restaurants contribute 22% of the regional GDP. Almost 66% of Bali's GDP originates from the DMA (BPS 2016). The 2016 Economic Census shows that the hotel business has generated the highest employment and second-largest annual output. Overall, tourism-related activities such as accommodation, food and beverages, and leisure share 33% of the total employment in the DMA.

The tourism development in Bali is inseparable from the island's open culture and attitude towards the outside world (Picard 2006, 21). Before the twentieth century, most area of Bali is controlled by several Hindu kingdoms, with the major economic activities revolving around slavery trade. Having conquered the island, the Dutch government began to build infrastructure and opened a shipping lane to Bali in the early twentieth century. This attracted a flow of tourism to Bali which continues to date. However, the mushrooming of hotels and accommodations did not start until the 1990s around leisure activities that concentrated in Badung Regency. The increase in tourist arrivals has made the DMA the main location for foreign investment in various accommodation facilities (Yuliandari, Martha, and Widodo 2014). In fact, half of the foreign direct investment in the DMA is related to global hotel chains. The massive development of tourism facilities contributed to a rapid rise in employment, especially in Denpasar City and Badung Regency, allowing the DMA to grow into a cosmopolitan area consisting of local migrants and many foreign residents. Denpasar and Kuta have been planned as the urban core, while other places such as Gianyar, Tabanan and Ubud are arranged as local centers for trade and services (Rahayu, Haigh, and Amaratunga 2018). The uncontrolled property growth in the DMA has motivated Bali's provincial government to introduce a moratorium on hotel development. However, due to its failure to prevent new massive development, the moratorium is considered ineffective (Yuliandari, Martha, and Widodo 2014).

Although the linkage between the three foregoing metropolitan areas to the global capital has begun since the Dutch occupation, each of them had different moments of

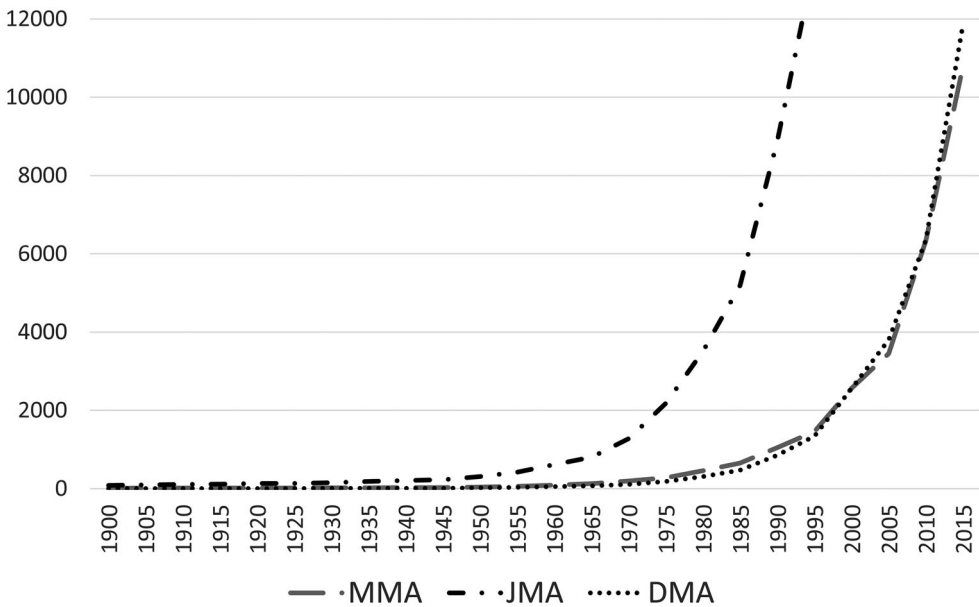


Figure 3. The growth of Medium and Large Enterprises in MMA, JMA, and DMA.

local economic leaps. By considering MLEs as long-term planned capital accumulation, the acceleration of the JMA's economy began two decades earlier than in the MMA and the DMA (Figure 3). The establishment numbers in the JMA have grown sharply since the Indonesian independence in 1945, while the growth in the MMA and DMA is strongly related to the policies of economic openness to foreign direct investment in the early 1970s. To date, there are just a few multi-national company branches in these two metropolises, as most foreign investment flows in indirectly through cooperation with national business entities using their names and positioning in Jakarta. Given the unequal endowment between regions, earlier government policies have kept development planning centralized and left most regional development initiatives up to local actors. This was the major cause of disparities during the early New Order era (Akita 1988).

Discussion

Kloosterman and Musterd (2001) argued that the reasons for urban restructuring could be summarized as economic, technological, or societal changes. However, they emphasized that research on polycentricity should not merely focus on physical or spatial form of an urban area and its functional relationship, but also on political entities and cultural identity. In other words, the context of development could offer a better understanding of urban spatial outcomes. Empirical evidence shows that most current polycentric urban regions such as San Francisco, Tuscany, Emilia Romagna, Mexico City, Barcelona, Tokyo, Beijing, and Hangzhou are on the scent of manufacturing development trajectory in their urban history (Burgalassi 2010; Cowell 2010; Fernández-Maldonado et al. 2014; Garcia-López and Muñiz 2010; Huang et al. 2017; Sorensen 2001; Yue, Liu, and Fan 2010). Other polycentric urban regions which primarily developed from the rise of

non-manufacturing industry, such as the Randstad in the Netherlands, are located in proximity to specialized cities, forming an urban network (Cowell 2010). Even if the manufacturing sector has shrunk or the economic structure has shifted towards services, manufacturing industry would still exist in the suburban area and be of significant importance.

The history of the three examined metropolitan areas portrays a long-term globalized urban development with different trajectories. Following Dick and Rimmer (1998), urban form divergences between the 1940s–1970s should have been re-converged after the 1980s due to the increase of trade and investment in global cities. Yet, the trade liberalization policy taken by the Indonesian government could not decrease the spatial concentration of industry. Industrial activities did not disperse to other regions because of the manufacturing industry's need to locate close to the main ports, decisions by policymakers and the situation of dictatorship and political instability (Sjöberg and Sjöholm 2004). Thus, the industry-led polycentric development argument seems to apply only to the JMA. The shift of the industries from the core of the JMA to the periphery triggered the post-suburbanization process featuring a fragmented and polycentric structure (Hudalah and Firman 2012). In fact, the recently formed subcenters in the JMA periphery specialize in manufacturing sectors (Sadewo, Syabri, and Pradono 2018).

Although the MMA has followed the same strategy by establishing the Medan Industrial Parks in 1988, its industrialization process seems to be going nowhere as the industrial land development trends stalled. The specialized location quotient of Deli Serdang's industry, for instance, indicates that the MMA has experienced a decentralization process. Yet, it seems too insufficient in scale to compete with the JMA because Deli Serdang as the main hinterland for the MMA could not escape its status as an underdeveloped region (Akhir et al. 2016). Meanwhile, the dependency on tourism activities has a double impact on urban development in the DMA. First, efforts to preserve the local culture have prevented the growth of manufacturing industries. Yet, this situation provides a conducive environment for the growth of cultural creative industries, which are less sensitive to distance. Second, the tourist attractions that are scattered around the other parts of Bali can mostly be reached within several hours from the current CBD of the DMA. This contributes to prevent the emergence of other subcenters away from the CBD.

The institutional factor plays an important role in steering these historical and economic developments of metropolitan areas and triggers their transformation into polycentric structures. Cowell (2010) argued that a polycentric region with a high level of economic complementarities would form under weak regional governments; in places with a strong regional identity; and in regions that have actively pursued polycentrism as a development strategy. The importance of planning efforts was found to be substantial in Tokyo in its satellite city development (Sorensen 2001) and in the industrial relocation and *hukou* (i.e. household registration) policy in China (Huang et al. 2017; Yue, Liu, and Fan 2010). Another important factor that encourages polycentric development is the availability of transportation infrastructure. The cases of Barcelona, Paris, Beijing, and Tokyo substantiate the importance of transportation system to encourage the employment relocation process (García-López and Muñiz 2010; García-López, Hémet, and Viladecans-Marsal 2017; Huang et al. 2017; Sorensen 2001). Under the influence of globalization, the factors of governance and accessibility could play a key role in achieving polycentric development (Sweet, Bullivant, and Kanaroglou 2016).

Although it was not the intention of the strategy, infrastructure biases may have also played an important role in the polycentric development of the JMA. The MLEs in the JMA were largely concentrated in the urban core until the extensive development of a toll road network facilitated its decentralization. As the national center of politics and economics, the JMA has become the center of infrastructure investments, thus surpassing other metropolitan areas in Indonesia. The JMA's toll road and rail networks cover a large area, allowing for the relocation of industry to its periphery. Furthermore, a diverse regional identity and complex institutional setting with many stakeholders have weakened the role of the local government. Industrial land development in the metropolitan outskirts is heavily controlled by the private sector (Hudalah and Firman 2012; Hudalah et al. 2013). The uncontrolled land development creates the contrast of planned and unplanned space in JMA's periphery, repeating what Winarso (2011) termed as "urban dualism".

The MMA and the DMA have relatively strong regional identities but less governmental support, as evident from their late infrastructure development. The Belawan-Medan-Tanjung Morawa toll road in the MMA was operational in 1986, two years before the Jakarta-Cikampek toll road. However, it took almost three decades for the government to expand the toll road coverage in the MMA. In the DMA, the Bali-Mandara toll road has only been operational since 2013 within a limited range from the airport to the inner city. While the MMA is already equipped with a passenger train system, the DMA has no train system at all. This lack of infrastructure makes it difficult to attract investment in manufacturing and encourage the establishment of new subcenters.

Conclusion

This paper contributes to the literature on polycentric urban development in Southeast Asia by describing the causes and consequences of polycentric development in three metropolitan areas in Indonesia, namely Medan, Jakarta, and Denpasar. We have selected these metropolitan areas because of their long historical engagement with global capitalism regardless of their striking differences. An individual examination of their subcenters gives insight into the current metropolitan spatial structure in Indonesia. While the employment centers in the JMA have spread to several locations in the periphery, we found only a single employment center in the MMA and DMA. This indicates that the economic activities in both metropolitan areas are still strongly concentrated. However, in contrast to the employment center of the MMA, which is located close to the traditional CBD, the employment center in the DMA is located outside the traditional CBD and the urban core. The contrast in CBD location findings implies that the DMA has different monocentric characteristics, compared to the MMA.

The supposition is confirmed by assessing the comparative polycentricity level of the three metropolitan areas; while the JMA and the MMA have a polarized morphology, the DMA's urban structure is in-between these two metropolitan areas. In addition, the commuter travel patterns reveal that the functional polycentricity of the MMA and DMA is focused on the core and the CBD. Although both regions have a similar functional polycentricity value, their in- and out-functional polycentricity are quite different. The DMA shows a more reversed commuting pattern to the supposedly suburban area while commuting in the MMA is dominated by the traditional pattern from suburban areas to the core. In the JMA, the multiple subcenters create a more balanced commuting

direction and the general functional polycentricity value indicates a moderate level of polycentricity. The variation of functional polycentricity between the three areas parallels the difference of their morphology; i.e. the location of CBD strongly corresponds to the commuting pattern. These findings highlight the need to incorporate both aspects in future polycentricity studies.

Regarding the specific contextual and historical development, this study argues that polycentricity is not necessarily dependent on population, nor economic size. What distinguishes the JMA and other established polycentric metropolises in the foreign countries, from the MMA and the DMA is the background of its economic development. Compared to agricultural or tourism-related industries, the manufacturing sector has proven to effectively boost urban transformations in Indonesia. Such difference in regional economic bases of urban development is likely to determine an urban form, but more evidence is needed to support this hypothesis. However, it should be noted that the existence of multiple subcenters from industrial complexes development in the JMA was not driven by the government's planning strategy. For years, manufacturing industries may have developed in the JMA because the region has been the focus of government investments. But industrial complex expansion and land conversion in the JMA's peripheral areas was largely controlled by private sectors. The provision of transportation infrastructure might help to pursue a desired urban form, yet, other regions seeking similar manufacturing development have failed to do so because national policies favored Jakarta as the capital and they lack the JMA's strategic location.

This study concludes that among the three Indonesian metropolitan areas that are the focus of this research, only the JMA has a polycentric spatial structure. This study confirms the previous findings of a polycentric structure in the JMA, but not in the other metropolitan areas. Despite using the latest data, this research has several shortcomings. First, the identification of urban morphology is limited to MLEs. Despite their significant contribution to employment, the total number of MLEs is less than the total amount of micro and small establishments. Thus, the conclusions about morphology are only part of the whole story. Second, the study only compared three metropolitan areas. For a more comprehensive conclusion, future studies should also compare Indonesian metropolises to similar-sized metropolises in other developing countries to control for more variation of urban development. Third, most studies in this field have analyzed the level of polycentricity (both functional and morphological) of the given city-regions, but relatively little is known about how a polycentric city-region has developed. Therefore, as Li and Phelps (2019) have indicated, future research could pay more attention to the combination of quantitative and qualitative empirical methods especially to explain the micro-macro level mechanisms behind the development of polycentric urban regions.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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